

Entity Relationship Analysis Report

E-Commerce Order Management Database System

Document Version: 1.0 | Focus: Relationship Architecture, Cardinality Analysis & Referential Integrity | Notation: Crow's Foot & Relational Algebra

1. Introduction & Conceptual Framework

This report provides an in-depth evaluation of the entity-relationship (ER) architecture for the **E-Commerce Order Management Database System**. While the Entity Analysis Report defined the structural attributes and constraints of individual entities, this analysis focuses exclusively on the **inter-entity connections, cardinality ratios, participation constraints, and referential integrity mechanisms** that govern data flow across the platform.

A robust relational data model relies on accurately translating real-world e-commerce business rules—such as checkout workflows, stock reservations, order splitting, and payment settlement—into mathematically sound relational structures. This document formalizes these interactions using Crow's Foot ER notation standards and defines the exact cascading behaviors required during schema implementation.

Core Architecture Summary

The system models **8 primary relationships** connecting **9 relational entities**. To maintain strict **3NF compliance** and eliminate data redundancy, all Many-to-Many (M:N) relationships have been decomposed into associative entities with composite structural dependencies.

2. Comprehensive Relationship Matrix

The following matrix consolidates all inter-entity relationships within the database schema, mapping parent (referenced) entities to child (referencing) entities alongside their cardinality and participation rules.

PARENT ENTITY	RELATIONSHIP	CHILD ENTITY	CARDINALITY	PARENT PARTICIPATION	CHILD PARTICIPATION
CUSTOMER	Places	ORDER	1 : N	Optional (0..N)	Mandatory (1..1)
CATEGORY	Contains	PRODUCT	1 : N	Optional (0..N)	Mandatory (1..1)
CATEGORY	Subcategorizes	CATEGORY	1 : N	Optional (0..N)	Optional (0..1)
PRODUCT	Tracked In	INVENTORY	1 : 1	Mandatory (1..1)	Mandatory (1..1)
ORDER	Includes	ORDER_ITEM	1 : N	Mandatory (1..N)	Mandatory (1..1)

PARENT ENTITY	RELATIONSHIP	CHILD ENTITY	CARDINALITY	PARENT PARTICIPATION	CHILD PARTICIPATION
PRODUCT	Listed In	ORDER_ITEM	1 : N	Optional (0..N)	Mandatory (1..1)
ORDER	Settled Via	PAYMENT	1 : N	Optional (0..N)	Mandatory (1..1)
ORDER	Fulfilled By	SHIPMENT	1 : N	Optional (0..N)	Mandatory (1..1)
CUSTOMER	Authors	REVIEW	1 : N	Optional (0..N)	Mandatory (1..1)
PRODUCT	Evaluated By	REVIEW	1 : N	Optional (0..N)	Mandatory (1..1)

3. Deep-Dive Cardinality & Business Rule Analysis

Each relationship is governed by explicit business rules that dictate how database records are created, linked, and maintained during platform operation.

3.1 Customer – Order Relationship

ONE-TO-MANY (1:N)

Business Rule: A registered customer can place zero orders (as a new browser) or multiple orders over their lifetime. Every order must be explicitly tied to an authenticated customer account. Guest checkouts are handled by creating a permanent or temporary customer profile first.

Crow's Foot Notation: CUSTOMER ||——<| ORDER

Referential Key Mapping: ORDER.CustomerID → CUSTOMER.CustomerID (NOT NULL)

3.2 Category – Product Relationship & Hierarchy

ONE-TO-MANY (1:N)

Business Rule: Product catalog organization requires every product to belong to exactly one leaf-level category. A category may house multiple products or remain empty during initial catalog setup. Categories can also self-reference to form an infinite depth taxonomy (e.g., Electronics → Computers → Laptops).

Crow's Foot Notation: CATEGORY ||——<0 PRODUCT

Referential Key Mapping: PRODUCT.CategoryID → CATEGORY.CategoryID (NOT NULL)

Recursive Mapping: CATEGORY.ParentCategoryID → CATEGORY.CategoryID (NULLABLE)

3.3 Product – Inventory Synchronization

ONE-TO-ONE (1:1)

Business Rule: To prevent stock discrepancies, each physical product SKU is assigned exactly one centralized inventory record. This record tracks on-hand stock, reserved stock (items in carts/pending checkouts), and warehouse location. When a new product is added, its inventory profile is automatically initialized.

Crow's Foot Notation: PRODUCT ||——|| INVENTORY

Referential Key Mapping: INVENTORY.ProductID → PRODUCT.ProductID (NOT NULL, UNIQUE)

3.4 Order – Payment Lifecycle

ONE-TO-MANY (1:N)

Business Rule: Although a typical order is settled via a single transaction, the schema enforces a 1:N cardinality. This architectural decision accommodates complex financial workflows including failed payment re-attempts, split-tender checkouts (e.g., Gift Card + Credit Card), and subsequent refund or chargeback transactions.

Crow's Foot Notation: ORDER ||——<0 PAYMENT

Referential Key Mapping: PAYMENT.OrderID → ORDER.OrderID (NOT NULL)

3.5 Order – Shipment Logistics Fulfillment

ONE-TO-MANY (1:N)

Business Rule: An order can be fulfilled through one or multiple physical shipments. Multi-shipment (1:N) support is mandatory for handling multi-warehouse order routing, backordered item dispatch, and bulky item deliveries. An order remains in 'Processing' status until all associated shipment records reach 'Delivered'.

Crow's Foot Notation: ORDER ||——<0 SHIPMENT

Referential Key Mapping: SHIPMENT.OrderID → ORDER.OrderID (NOT NULL)

4. Associative Entity Resolution (M:N Decomposition)

Relational database engines cannot directly implement Many-to-Many (M:N) relationships without introducing data redundancy and violating normalization rules. In this schema, two major M:N relationships have been structurally resolved into associative entities.

4.1 Order – Product Resolution via ORDER_ITEM

An order can contain multiple products, and a single product can be purchased across thousands of orders over time. This M:N relationship is decomposed into two 1:N relationships via the ORDER_ITEM associative table.

- **Historical Price Freezing:** ORDER_ITEM captures UnitPriceAtPurchase. If a product's price changes in the catalog tomorrow, historical order records remain mathematically accurate.
- **Granular Quantity & Line Total:** Stores exact quantities purchased per item and computes line-item subtotals independently of the parent order total.
- **Composite Uniqueness:** A composite unique constraint on (OrderID, ProductID) ensures that the same product is not listed twice as separate rows in the same order (quantities are merged instead).

4.2 Customer – Product Resolution via REVIEW

Customers evaluate multiple products, and products receive feedback from multiple buyers. The REVIEW entity acts as an associative bridge between CUSTOMER and PRODUCT.

- **Verified Buyer Enforcement:** Application logic cross-references ORDER_ITEM to ensure that a customer can only author a review for a product ID they have successfully purchased.
- **Feedback Metadata:** Encapsulates rating scores (1-5 stars), text commentary, timestamps, and moderation flags within a clean transactional record.

5. Referential Integrity & Cascading Action Matrix

To guarantee database consistency and prevent orphaned records during UPDATE or DELETE operations, strict cascading rules must be embedded within the Foreign Key constraint definitions.

FOREIGN KEY LINK	ON DELETE ACTION	ON UPDATE ACTION	ARCHITECTURAL JUSTIFICATION
ORDER.CustomerID	RESTRICT / NO ACTION	CASCADE	Prevents deletion of customer accounts that have historical order financial records. Prevents accounting anomalies.
PRODUCT.CategoryID	RESTRICT / NO ACTION	CASCADE	Prevents accidental deletion of categories containing active catalog products.
INVENTORY.ProductID	CASCADE	CASCADE	If a product is permanently removed from the catalog, its corresponding stock tracking profile is deleted automatically.
ORDER_ITEM.OrderID	CASCADE	CASCADE	If an order is deleted (e.g., draft checkout purged), all associated line items must be removed to prevent orphaned data.
ORDER_ITEM.ProductID	RESTRICT / NO ACTION	CASCADE	Products included in historical orders cannot be deleted from the system; they must be marked 'Discontinued' instead.
PAYMENT.OrderID	RESTRICT / NO ACTION	CASCADE	Financial transaction logs must be preserved permanently for audit and compliance purposes.
SHIPMENT.OrderID	CASCADE	CASCADE	Shipment tracking schedules are tied directly to the lifecycle of the parent order transaction.
REVIEW.ProductID	CASCADE	CASCADE	If a product is completely purged, associated customer reviews are cleaned up automatically.
REVIEW.CustomerID	SET NULL / CASCADE	CASCADE	If a customer closes their account, reviews can be anonymized (SET NULL) or removed based on privacy compliance (GDPR).

Week 2 Relationship Analysis Sign-Off

This report completes the relationship and cardinality analysis required for Week 2. Together with the **Entity Analysis Report**, the conceptual and logical data models are fully defined and validated. The project is ready to progress to Week 3: SQL Data Definition Language (DDL) Scripting and Schema Implementation.